

AIRBNB PRICE PREDICTION

End-to-End Machine Learning Project

DS605

Fundamentals of Machine Learning

LAB ASSIGNMENT 4

Regression + Streamlit

A practical workflow from raw Airbnb data to a working price prediction application.

48,895
original listings

16
original columns

0.5189
final test R^2

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Dataset: AB_NYC_2019.csv

01 / PROJECT SNAPSHOT

The goal of this project was to build a complete machine learning workflow for estimating the nightly price of New York City Airbnb listings. The work covers data cleaning, exploratory analysis, feature engineering, model comparison, tuning, evaluation and saving the final pipeline for a Streamlit app.

Dataset	AB_NYC_2019.csv
Original size	48,895 rows × 16 columns
After cleaning	48,410 rows × 16 columns
Train / test split	38,728 training / 9,682 testing
Target	price
Models compared	Linear Regression, Ridge, Random Forest
Final model	Tuned Random Forest
Saved pipeline	models/airbnb_price_pipeline.pkl

PROJECT FLOW



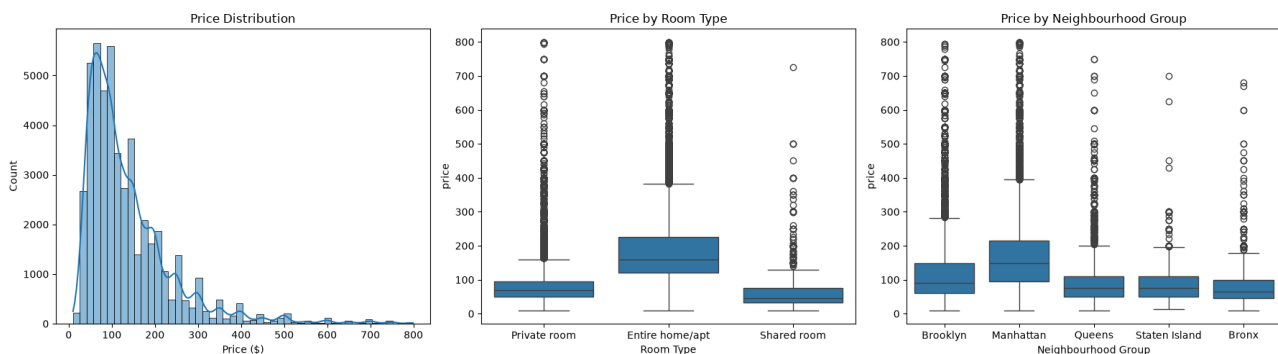
02 / TASK 1 — DATA ANALYSIS

I first checked the dataset size and missing values, then cleaned the target variable before moving to exploratory analysis. Duplicate rows were removed, zero-price listings were excluded, and prices above the 99th percentile were removed to reduce the effect of extreme values.

Check	Result
Original rows	48,895
Original columns	16
Final rows	48,410
Zero-price listings	Removed
99th percentile price limit	\$799
Missing-value handling	Pipeline imputation

WHAT THE EDA SHOWED

Room type and location showed clear differences in price. The median price was highest for entire homes/apartments (\$160), followed by private rooms (\$70) and shared rooms (\$45).



The analysis also showed that the price distribution is strongly right-skewed. This supported the use of log-based numerical features in the modelling workflow.

03 / TASK 1 — FEATURE PREPARATION

The model does not need listing IDs, host names or review dates as direct predictors. These fields were removed. I also created a few simple numerical features to make the input more useful for the regression models.

Feature	Why it was used
neighbourhood_group	Original input
neighbourhood	Original input
latitude	Original input
longitude	Original input
room_type	Original input
minimum_nights	Original input
number_of_reviews	Original input
reviews_per_month	Original input
calculated_host_listings_count	Original input
availability_365	Original input
reviews_per_availability	Engineered feature
minimum_nights_log	Engineered feature
reviews_log	Engineered feature
availability_log	Engineered feature

PREPROCESSING PIPELINE

Numerical columns were filled using the median and standardized. Categorical columns were filled using the most frequent value and one-hot encoded. The complete preprocessing workflow was kept inside the same scikit-learn Pipeline as the model, so the saved file can be reused by the application.

04 / TASK 2 — MODEL COMPARISON

Three regression models were compared on the same train/test split. MAE and RMSE measure prediction error, where lower is better. R^2 shows how much of the variation in price is explained by the model, where higher is better.

Model	MAE	RMSE	R^2
Random Forest	44.85	72.94	0.5111
Ridge Regression	49.47	78.78	0.4297
Linear Regression	49.53	78.79	0.4295

BEST BASELINE

Random Forest performed best among the three baseline models, with the lowest MAE and RMSE and the highest R^2 .

HYPERPARAMETER TUNING

Parameter	Selected value
n_estimators	200
max_depth	15
min_samples_split	5

GridSearchCV was used with 3-fold cross-validation and a small parameter grid. The selected Random Forest used 200 trees, maximum depth 15 and minimum samples split of 5.

05 / FINAL MODEL

The tuned Random Forest was used as the final model in the notebook. Its test performance was measured after fitting the complete preprocessing + model pipeline.

44.25 MAE	72.36 RMSE	0.5189 R ²
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Metric	Training	Testing
RMSE	53.14	72.36
R ²	0.7371	0.5189

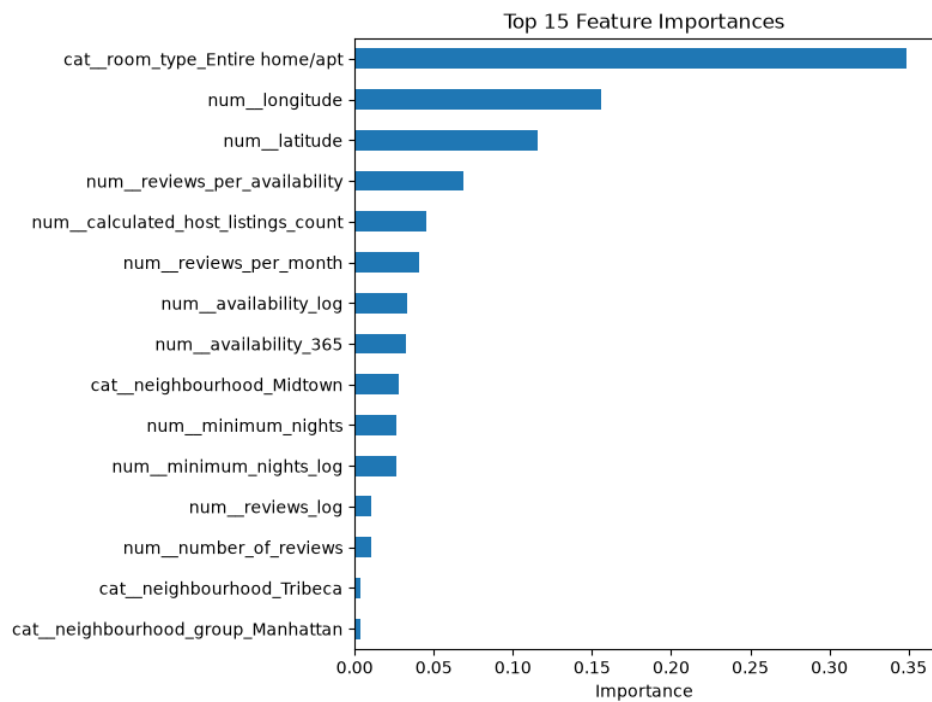
The training scores are better than the test scores, which is expected for a model that learns from the training data. The gap is noticeable but the test R² of 0.5189 shows that the selected features still provide useful predictive information.



06 / WHAT THE MODEL LEARNED

The Random Forest feature importance output gives a simple view of which processed features were most influential in the fitted model.

Feature	Importance
Entire home/apt	0.3487
Longitude	0.1561
Latitude	0.1160
Reviews / availability	0.0687
Host listings count	0.0452
Reviews per month	0.0407



07 / STREAMLIT APPLICATION

A Streamlit application was created around the saved pipeline. The interface collects listing information such as neighbourhood, room type, coordinates, minimum nights, reviews and availability, then creates the same engineered features used during training before making a prediction.

Location	Neighbourhood group, neighbourhood, latitude, longitude
Property	Room type
Stay	Minimum nights, availability
Reviews	Number of reviews, reviews per month
Host	Host listings count

RUNNING THE APPLICATION

The project can be run locally after installing the packages listed in requirements.txt. The Streamlit entry point is app/airbnb_predictor.py and the saved model is loaded from models/airbnb_price_pipeline.pkl.

MODEL FILE

The final preprocessing and Random Forest pipeline was saved with Joblib. This keeps the preprocessing steps together with the trained estimator and makes it easier to reuse the same workflow in the application.

08 / CONCLUSION & LIMITATIONS

The project completed an end-to-end machine learning workflow for Airbnb price prediction. The data was cleaned, explored and transformed, three regression models were compared, Random Forest was tuned, and the final pipeline was saved for use in a Streamlit application.

MAIN TAKEAWAYS

- Room type and location show clear differences in Airbnb prices.
- Random Forest gave the strongest baseline performance among the tested models.
- The tuned final model achieved a test MAE of 44.25 and R^2 of 0.5189.
- The complete preprocessing and model workflow was saved with Joblib.
- The Streamlit app provides a simple interface for making new predictions.

LIMITATIONS

- The dataset is from 2019, so it may not reflect current Airbnb prices.
- The model does not include every possible pricing factor, such as amenities, seasonality or current demand.
- The model still leaves a meaningful amount of price variation unexplained.
- Predictions should be treated as estimates rather than exact market prices.

Final model: Random Forest Regressor

Test MAE: 44.25 **Test RMSE:** 72.36 **Test R^2 :** 0.5189

Deliverables: analysis notebook • saved model pipeline • Streamlit application • requirements.txt • project documentation

LIVE STREAMLIT APPLICATION

The trained Airbnb price prediction model has been deployed as an interactive web application.

Application	Airbnb Price Predictor
Platform	Streamlit Community Cloud
Model	Tuned Random Forest Regression
Dataset	New York City Airbnb Open Data 2019
Status	Public deployment

Public App Link

<https://airbnbratepredictor.streamlit.app/>

The application accepts Airbnb listing details such as neighbourhood, room type, location, minimum nights, reviews, host listing count, and availability, and returns an estimated nightly price.

Deployment note: The application is hosted online so it can be opened and tested without running Streamlit locally.

This page was added to the final project report after successful deployment of the Streamlit application.